10TH CLASS BIOLOGY — NEW SYLLABUS (PECTAA)

Chapter 5 — Coordination

Solved Exercise | New Book (PECTAA)

Exercise (Solved)

A. Select the correct answers for the following questions.

1. Schwann cells are the supporting cells for neurons. They make:

(a) Cell body (b) Axon
(c) Myelin sheath (d) Dendrites

Answer: (c) — Schwann cells are a type of neuroglia, and in many neurons they produce the insulating covering called the myelin sheath over the membrane of the axon. The cell body, dendrites and axon are parts of the neuron itself.

2. Which part of the brain controls balance while riding a bicycle?

(a) Cerebrum (b) Medulla
(c) Cerebellum (d) Spinal cord

Answer: (c) — The cerebellum is the second largest part of the brain and its surface is highly folded. It coordinates balance and muscle movements, so it is the part that keeps us balanced while riding a bicycle.

3. Which part is responsible for sensations of temperature, hunger, thirst, and sleep?

(a) Hypothalamus (b) Thalamus
(c) Pons (d) Cerebellum

Answer: (a) — The hypothalamus lies above the midbrain and just below the thalamus, and it is about the size of an almond. It regulates body temperature, hunger, thirst, sleep and emotional states, and it also controls the secretions of the pituitary gland.

4. If a patient cannot remember things, which part of their brain may be affected?

(a) Cerebrum (b) Cerebellum
(c) Medulla (d) Spinal cord

Answer: (a) — The temporal lobe of the cerebrum contains the regions which deal with memory, learning, feelings and hearing. So if a person cannot remember things, the cerebrum (especially the temporal lobe) may be affected.

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5. During danger, which system activates quickly?

(a) Parasympathetic (b) Skeletal
(c) Endocrine (d) Sympathetic

Answer: (d) — The sympathetic nervous system prepares the body in emergency situations; this is called the “fight or flight” response. It dilates the pupils, speeds up the heartbeat, increases the breathing rate and inhibits digestion.

6. Why are reflex actions faster than normal actions?

(a) Brain processes them quickly (b) Brain is not involved
(c) Signals move slower (d) They need thinking

Answer: (b) — Reflex actions are quick involuntary responses in which the brain is not involved. The impulse travels only through the spinal cord along the reflex arc, so the response is produced without any thinking and is therefore faster.

7. Which of these acts as exocrine as well as endocrine gland?

(a) Pituitary (b) Thyroid
(c) Adrenal (d) Pancreas

Answer: (d) — The pancreas is a part of the digestive system, where it works as a ducted (exocrine) gland and secretes digestive enzymes. Its islets of Langerhans work as an endocrine gland and secrete insulin and glucagon into the blood.

8. The chemical messengers of the endocrine system are called:

(a) Neurotransmitters (b) Hormones
(c) Neurons (d) Enzymes

Answer: (b) — Hormones are the chemicals released from the endocrine glands which regulate different body functions. Neurotransmitters carry messages across a synapse, which is a function of the nervous system.

9. What could happen if hormone of parathyroid gland were missing?

(a) Low calcium levels (b) High blood sugar
(c) Weak muscles (d) Poor digestion

Answer: (a) — Parathormone stimulates the transfer of calcium ions from the bones to the blood, so it increases the calcium level of the blood. If this hormone is missing, the blood calcium level falls, which leads to tetany (muscle twitching, cramps and convulsions).

10. Which endocrine gland prepares the body for “fight or flight” response?

(a) Thyroid (b) Pituitary
(c) Adrenal (d) Pancreas

Answer: (c) — The adrenal medulla produces epinephrine (adrenaline) and nor-epinephrine, which prepare the body for the “fight or flight” response in an emergency by increasing the heart rate, blood pressure, blood glucose level and blood flow to the heart and lungs.

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B. Write short answers.

1. Define stimulus with examples.

Any change in the external or internal environment which can initiate a response in the body is called a stimulus. Touch, light, sound, heat, cold, pressure and infection are examples of stimuli. The organ, tissue or cell which receives a stimulus is called the receptor; for example, the ears are the receptors for the stimulus of sound.

2. Differentiate between sensory, motor, and inter-neurons.

Sensory neurons	Inter-neurons	Motor neurons
They transmit nerve impulses from the receptors to the CNS.	They are present in the brain and the spinal cord.	They transmit nerve impulses from the inter-neurons to the effectors.
Their dendrites are connected with the receptors.	They receive impulses from sensory neurons and pass them to motor neurons.	Their axons are connected with the muscles or the glands.
They carry information about a stimulus towards the centre.	They process and connect the information inside the CNS.	They carry the order of the centre to the effector, which then produces the response.

3. Define reflex action and the reflex arc.

Reflex action: reflex actions are the quick, involuntary responses in which the brain is not involved. They are controlled by a reflex arc and protect the body from harm; for example, we pull our hand away from a hot flame without thinking about it.

Reflex arc: a reflex arc is the pathway of neurons over which the nerve impulses travel during a reflex action. Its pathway is: receptor → sensory neuron → inter-neuron of the spinal cord → motor neuron → effector.

4. Differentiate between the following: (a) Dendrite and axon

Dendrite	Axon
It is a small branch which projects from the cell body.	It is a long branch which arises from the cell body.
There are many dendrites on one neuron.	There is usually only one axon on a neuron.
It is short and highly branched at its end.	It is long, and its terminal end is also branched.
It transmits nerve impulses towards the cell body.	It transmits nerve impulses away from the cell body, towards other neurons, muscles or glands.

(b) Cerebrum and Cerebellum

Cerebrum	Cerebellum
It is the largest part of the forebrain and of the whole brain.	It is the second largest part of the brain.
It is present in the upper part of the brain and is divided into two cerebral hemispheres.	It lies behind the medulla oblongata.
Its upper layer, the cerebral cortex, is highly folded and is made of grey matter.	Its surface is also highly folded.
It is the centre of thinking, memory, learning, speech, feelings, vision and voluntary movements.	It coordinates balance and muscle movements.

(c) Thalamus and Hypothalamus

Thalamus	Hypothalamus
It is a structure wrapped by the cerebrum.	It lies above the midbrain and just below the thalamus, and is about the size of an almond.
It conveys information coming from the receptors to the cerebrum.	It regulates body temperature, hunger, thirst, sleep and emotional states.
It is involved in pain perception and consciousness.	It controls the secretions of the major endocrine gland, the pituitary gland.

5. Identify the diagram and label the different parts.

The diagram shows a model neuron (motor neuron). The parts to be labelled are: the cell body with its nucleus, the dendrites, the axon, the myelin sheath formed by the Schwann cells, the nodes of Ranvier between the Schwann cells, and the branched terminal end of the axon. The arrow shows the direction of the nerve impulse, which travels from the dendrites towards the cell body and then along the axon.

6. Identify the following diagram, and locate the white matter, grey matter, three types of neuron, and ganglia.

The diagram shows a cross section of the spinal cord with the spinal nerves. The **grey matter** is the butterfly-shaped central region, and the **white matter** is the outer region of the spinal cord. The **sensory neuron** enters through the dorsal root, which contains a swelling called the **ganglion** where the cell bodies of the sensory neurons are present. The **inter-neuron** lies inside the grey matter of the spinal cord, and the **motor neuron** leaves the cord through the ventral root. In this way the diagram also shows the reflex arc.

C. Write answers in detail.

1. Describe the structure of a neuron and support your answer with a labelled diagram.

Neurons or nerve cells are the cells which transmit impulses in the body in the form of nerve impulses. A nerve impulse is a wave of electrochemical change that travels across the membrane of a neuron. A neuron consists of three major parts: the cell body, the dendrites and the axon.

Cell body: it contains the nucleus, other organelles and most of the cytoplasm.

Dendrites: these are small branches which project from the cell body. They receive nerve impulses and transmit them towards the cell body, so they bring information to the neuron.

Axon: it is a long branch arising from the cell body, and its terminal end is further branched. It transmits the nerve impulses from the cell body to other neurons, muscles or glands.

Myelin sheath: neurons are supported by specialised cells called neuroglia, and Schwann cells are a type of neuroglia. In many neurons the Schwann cells produce an insulating covering called the myelin sheath over the membrane of the axon. Axons covered by the myelin sheath are called myelinated and those without it are non-myelinated. The myelin sheath is interrupted at intervals by small gaps called the nodes of Ranvier, and the nerve impulse jumps from node to node, which increases the speed of the impulse.

Labelled diagram: the diagram should show dendrites, the cell body with the nucleus, the axon with the myelin sheath, the Schwann cells, the nodes of Ranvier, and the branched axon endings, with an arrow showing the direction of the nerve impulse.

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2. What are the differences and similarities among the three types of neurons?

Similarities: all three types are nerve cells built in the same way, with a cell body, dendrites and an axon. All of them carry the message in the form of a nerve impulse, all of them have a myelin sheath on many of their axons, and all of them work together in a reflex arc or in the normal pathways of the nervous system.

Feature	Sensory neuron	Inter-neuron	Motor neuron
Direction of impulse	From receptor to CNS	Within the CNS	From CNS to effector
Position	Its cell body lies in the ganglion of the dorsal root of the spinal nerve.	It lies completely inside the brain or the spinal cord.	Its cell body lies in the grey matter of the CNS and its axon goes out through the ventral root.
Number	Countless in the body, one for each receptor pathway.	The largest number, because most of the neurons of the CNS are inter-neurons.	Countless in the body, ending at muscles and glands.

3. Write the major functions of the forebrain, midbrain, and hindbrain.

Forebrain: it is the largest area of the brain and consists of the cerebrum, the thalamus and the hypothalamus. The cerebrum is the centre of thinking, problem solving, judgment, speech, memory, learning, feelings, hearing, vision and voluntary movements. The thalamus conveys information coming from the receptors to the cerebrum and is involved in pain perception and consciousness. The hypothalamus regulates body temperature, hunger, thirst, sleep and emotional states, and controls the secretions of the pituitary gland.

Midbrain: it is located between the forebrain and the hindbrain, so it acts as a bridge between them and conveys information between them. It also enables some reflexes related to vision and hearing.

Hindbrain: it consists of the medulla oblongata, the cerebellum and the pons. The medulla oblongata transmits impulses between the spinal cord and the higher parts of the brain, controls autonomic functions such as heartbeat, breathing and blood pressure, and controls reflexes such as vomiting, coughing and sneezing. The cerebellum coordinates balance and muscle movements, and the pons makes connections between the spinal cord and the cerebellum and between the cerebrum and the cerebellum, and also regulates alertness, sleep and wakefulness.

4. What is meant by reflex action? Describe an example of reflex action and identify the reflex arc in this example.

Reflex actions are the quick, involuntary responses in which the brain is not involved, and they are controlled by a reflex arc. A reflex arc is the pathway of neurons over which the nerve impulses travel during a reflex action. Reflex actions protect us from harm, for example we pull our hand away from a hot flame without thinking about it. This happens in the following way:

(i) The skin receptors detect the stimulus of heat.

(ii) The sensory neuron sends the nerve impulse to the inter-neuron of the spinal cord.

(iii) The inter-neuron transmits the nerve impulse to the motor neuron.

(iv) The motor neuron carries the nerve impulse to the effector, which is the muscle of the arm.

(v) The effector produces the response by contracting, so our hand is pulled away.

In this example the reflex arc is: skin receptors → sensory neuron → inter-neuron of the spinal cord → motor neuron → muscles of the arm. Because the message is handled in the spinal cord and does not go to the brain, the response is very quick.

5. State the hormones of the pituitary gland and write their functions.

The pituitary gland is a small gland present in the brain, suspended from the hypothalamus by a short stalk. It has two lobes, the anterior lobe and the posterior lobe.

Lobe	Hormone	Function
Anterior	Growth hormone (GH)	Stimulates the growth of muscle, bone and other tissues. Excess causes gigantism and deficiency causes dwarfism.
Anterior	Adreno-cortico-tropic hormone (ACTH)	Stimulates the adrenal cortex to produce its hormones, which control stress.
Anterior	Thyroid-stimulating hormone (TSH)	Stimulates the thyroid gland to produce thyroxine, which controls metabolism.
Anterior	Luteinizing hormone (LH)	Needed for the release of egg cells from the ovary (ovulation) in females; in males it stimulates the testes to produce sex hormones.
Anterior	Follicle-stimulating hormone (FSH)	Involved in the reproductive system, where it helps in the maturation of gametes.

Lobe	Hormone	Function
Posterior	Antidiuretic hormone (ADH) or vasopressin	Increases the reabsorption of water from the nephrons of the kidney, so it controls blood volume and blood pressure.
Posterior	Oxytocin	Initiates contractions of the uterus during childbirth and stimulates the flow of milk during breastfeeding.

6. Write notes on adrenal glands, pancreas, ovaries and testes.

Adrenal glands: an adrenal gland is located above each kidney. Each gland has an inner core, the medulla, and an outer layer, the cortex. The adrenal medulla produces epinephrine (adrenaline) and nor-epinephrine, which take part in the emergency response of the sympathetic nervous system and prepare the body for the “fight or flight” response; they increase the heart rate, blood pressure, blood glucose level and blood flow to the heart and lungs. The adrenal cortex secretes cortisol, which promotes the production of glucose from proteins, and aldosterone, which raises blood pressure and volume by stimulating salt absorption and water retention by the kidneys.

Pancreas: the pancreas has two functions. It is a part of the digestive system, where it works as a ducted (exocrine) gland and secretes digestive enzymes. It also has special cells called the islets of Langerhans, which work as an endocrine gland and secrete insulin and glucagon. Insulin lowers the blood glucose level by stimulating the body cells to store glucose or use it for energy, while glucagon stimulates the release of glucose from the liver into the blood.

Ovaries: the ovaries are the female gonads. They produce the egg cells and secrete the hormones oestrogen and progesterone, which cause the release of the egg from the ovary and regulate the female secondary sex characteristics, such as the development of the breasts.

Testes: the testes are the male gonads. They produce the sperms and secrete a group of sex hormones called androgens, for example testosterone. Testosterone regulates the male secondary sex characteristics, such as the growth of hair on the face and the coarseness of the voice.

D. Inquisitive questions

1. Why is the spinal cord important in reflex actions?

The spinal cord is the coordinating centre for many simple reflexes. In a reflex action the message does not have to go to the brain; the sensory neuron brings the impulse to the spinal cord, an inter-neuron present there passes it to the motor neuron, and the motor neuron takes it to the effector. Because this pathway is short, the response comes very quickly and the body is saved from harm, for example when we touch something hot. The spinal cord also sends a copy of the message to the brain so that we become aware of the pain afterwards. If the spinal cord is damaged, such protective reflexes cannot take place.

2. Why is adrenaline called an “emergency hormone”?

Adrenaline (epinephrine) is secreted by the adrenal medulla when a person faces an emergency or danger, so it is called the emergency hormone. It prepares the body for the “fight or flight” response by increasing the heart rate, the blood pressure, the blood glucose level and the blood flow to the heart and lungs. It also enlarges the bronchioles so that more air can be taken in, and dilates the pupils so that vision becomes clearer. In this way the body gets more energy and oxygen within seconds, and the person can either fight the danger or run away from it.

3. If pancreas releases too much insulin, what happens to blood sugar?

Insulin lowers the blood glucose level by stimulating the body cells to store glucose or to use it for energy. If the pancreas releases too much insulin, the glucose is removed from the blood very quickly, so the blood sugar level falls below normal. This condition is called hypoglycaemia. The person feels weakness, sweating, shivering, hunger, dizziness and confusion, and in severe cases he may faint or go into a coma. Such a patient is given something sweet immediately, such as sugar or glucose water.

4. If the medulla of a patient is injured, which function is most at risk?

If the medulla oblongata is injured, the autonomic functions controlled by it are most at risk, especially breathing and the heartbeat. The medulla controls the heartbeat, the heart rate, breathing and blood pressure, and also the reflexes of vomiting, coughing and sneezing. Since these functions go on without our will, an injury to the medulla can stop breathing or make the heartbeat irregular, which is immediately dangerous for life. That is why this part of the brain is protected inside the skull together with the rest of the brain.

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